

Post-Detection report for 2023 IAA SETI Permanent Committee meeting, 3 Oct 2023, Baku / online

Report given remotely by Kathryn Denning. (Associate Professor, Anthropology, York University, IAA SETI Committee member since 2005, SETI Institute Science Advisory Board member). kdenning@yorku.ca

Prefatory note

As in my last update, we do not currently have a formally organized Post-Detection Working Group within the Committee, but (a) some members and other researchers from outside the committee have been busily working on post-detection-related topics, and also (b) we have had a small task group doing preliminary work specifically on the Protocols, *i.e.* starting the groundwork for a revision.

Previous post-detection updates are available [here](#) (2022) and [here](#) (2021): much of what was said there is still generally applicable. Below I provide:

- a brief summary of *some* of the developments since our last meeting in Sept 2022 (N.b. this summary is not comprehensive, and I would be glad to learn of anything I missed)
- a specific update on the Protocols task group's work so far (work in progress)

A few highlights/themes:

- strong interest in SETI “discovery and society” topics, including post-detection and societal impact, continues, both within and beyond the Committee
- the Committee has much work ahead concerning Protocols updates and related matters, *e.g.* best practices in relation to post-detection in today's information environment, given new and emerging search strategies and potential detection types, and the expanding community of searchers – all of which are vastly different now from when the Protocols were first formulated (shortly before the emergence of the WWW), and even from when they were last updated (2010)
- addressing these issues comprehensively will require background research, including the integration of expertise which is currently not well-represented within the Committee itself
- it bears repeating that post-detection related research is [essential](#), but persistently underfunded
- outbound transmissions from Earth – whether *de novo* or reply – continue to be a subject of debate, and at times friction, in this research community, and constructive ways forward are needed
- the full continuum of potential technosignatures, how they should be approached scientifically, and which organizations should be addressing different phenomena, continues to be a subject of debate and discussion (*e.g.* [NASA UAP study](#)), although UAPS are generally outside the remit of our own committee
- discussions in the biosigs/technosigs community about standards of evidence, confidence in life detection, scientific consensus, and communication strategies continue to develop, and do intersect with technosignatures
- mis/disinformation online is a serious and escalating concern which we need to address to the extent possible, *e.g.* with proactive planning for resilient information dissemination strategies which could rise to the challenges of a real discovery event

Overview of today's update

1. Papers on themes related to Post-detection in 2023 IACs
2. Some relevant recent and upcoming activities of SETI Permanent Committee members, including new research groups, projects, talk series, conferences and symposia with post-detection relevance
3. A few noteworthy publications (just examples...)
4. Protocols Revisions: work in progress

1. Post-detection Relevant Abstracts from IAC 2023

from <https://iafastro.directory/iac/browse/IAC-23/A4/>

As always, all the papers in the SETI sessions, including the technical ones, are of relevance to post-detection, because they illustrate or explore the forms of detection which are now possible or may imminently be possible. Some contributions which more specifically focus on topics related to public reaction and related policy etc. include the following. (N.b. some of the talks listed below were withdrawn/not presented, but the abstracts are still of interest.)

- **Lori Walton:** "Billingham Cutting Edge Lecture" - The History of the IAA SETI Permanent Committee - 1990 to 1999
- **Kate Genevieve *et al*:** The SETI Post-Detection Hub: Preparing Humanity for Contact
- **Carol Oliver *et al*** (presented by Les Tennen): Revising the SETI Post-Detection Protocols for the 2020s and Beyond: A Report on Work in Progress
- **Kate Genevieve *et al*:** The ocean calls - SETI, lunar astronomy and scenarios at the 21-cm hydrogen line
- **Daniela de Paulis:** Composition, decoding and interpretation of the simulated extraterrestrial message in the interdisciplinary project "A Sign in Space".
- **Paolo Musso *et al*:** Astrobiology and SETI in Peru: recent and future advances
- **Paolo Musso *et al*:** Insights for SETI from linguistics studies in the Peruvian Amazon
- **Gabriela Radulescu:** Regulating Outer Space: Contact with Extraterrestrial Intelligence in the International Astronautical Community in the late 1950s
- **Debarshi Mukherjee:** Societal Implications of Extraterrestrial Intelligence: Public Reaction, Risk Communication, and Possible Impacts on Society
- **Ugur Guven:** How to Distinguish a Potential Spaceship in Search for Extraterrestrial Intelligence?
- **Suresh Bhattarai *et al*:** Are we alone? People's perception on an ExoWorld, its planet and possible life on it

2. Some relevant recent and upcoming activities of SETI Permanent Committee members

- The SETI Institute and Breakthrough Listen continue their extensive public outreach efforts.
- Some other SETI projects also have a significant public-interaction component (will work on a list for next time! Please do contact me with details.)
- On the technical side, Breakthrough Listen and the SETI Institute are working on formalizing access to telescope resources for technosignature candidate follow-up, including dedicated resources like the Allen Telescope Array and community-scheduled resources such as the Very Long Baseline Array. (Andrew Siemion can provide more details.)
- [SETI Post-Detection Hub](#) at St. Andrews (Coordinator John Elliott will report additional details). The Hub, initially a UKSRN initiative, launched in November 2022 and is jointly hosted by the [St Andrews Centre for Exoplanet Science](#) (by Martin Dominik and Emily Finer) and the [Centre for Global Law and Governance](#), with additional affiliates Durham Law School, Manchester University, etc.. The Hub now involves several dozen interdisciplinary researchers, some who have been working on SETI topics for years and decades, and others who are new to SETI/technosignatures but have significant relevant expertise e.g. in science communication research, public policy, law, etc. Multiple SETI PC members are participants. Stay tuned for more as the Hub projects evolve. N.b. It is important to integrate our conversations in the science + interdisciplinary research communities as much as possible.
- **Breakthrough Discuss 2023:** The rapid expansion of technosignatures searches and solar system exploration, presented by a variety of speakers including SETI PC members, as always poses some new challenges for post-detection / discovery & society, although the latter are not usually explored in depth in the conference. [Talk Playlist](#).
- **PSETI (Penn State):**

The [PSETI seminar series](#) run by Jason Wright and colleagues includes SETI & Society / post-detection topics, and recordings are available.

The [PSETI 2023 symposium](#) included “SETI & Society” topics and post-detection, in the [plenary](#), [posters](#), and [breakouts](#).

- Talk recordings are available [here](#). Talks were primarily technical, but also included a review talk from K Denning ([Post-Detection for the 2020s: Navigating Discovery](#)) and a contribution on SETI history from [Gabriela Radulescu](#).
- N.b. the technical talks and the NASA Technosignature Gap Study (presented by Nick Siegler, Virisha Timmaraju, Rich Terrile) highlight, again, the rapid expansion of possible discovery scenarios.
- A breakout session, [“The Alien in the Machine: Post-Detection in the Era of AI”](#), was run by K Denning, Anamaria Berea, Rebecca Charbonneau, Steven J Dick, Chelsea Haramia, Brian S McConnell, & George Profitiliotis. Please see that link for the breakout session talk

abstracts/program, which included a group exercise run by foresight expert G. Profitiliotis. (Partial recording to come.)

- N.b. A substantial Google Drive folder of post-detection resources was created for participants in last year's 2022 [Post-Detection breakout session](#). It includes many papers from the PC's history, and some current resources. PC members who would like access can [click here](#) and request it, or email kdenning@yorku.ca
- **Daniela de Paulis and team, [A Sign in Space](#) (Daniela will report more details).** This project, presented by PC member Daniela de Paulis in 2021 and 2022 as work in progress, proceeded in 2023 and has been a great success. Preliminary stages involved considerable discussion with an advisory group regarding post-detection issues and the public, and the project itself involved many people in the SETI astronomy and general SETI community, as well as a multitude of public participants. De Paulis is also a [SETI Institute Artist in Residence](#).
- **[The Technosignatures Seminar Series](#)** at NASA Goddard, continues to feature speakers on a variety of technosignatures topics, including some from SETI PC members, and some directly relevant to post-detection. The website has a past and future speaker list and recordings of presentations. To join the mailing list, contact our member Ravi Kopparapu: ravikumar.kopparapu@nasa.gov
- **Confidence in Life Detection (CoLD) scale and Standards of Evidence for Biosignatures ... continued...** In the 2022 report, I noted that substantial discussion had occurred in 2021-22 regarding the CoLD scale and Standards of Evidence for Biosignatures, involving some members of the technosignatures community.

Those discussions continue in the biosig/technosig community, and are being extended into more interdisciplinary work about communicating discoveries: e.g. the NASA Astrobiology Program is running a virtual workshop in 2024, "[Communicating Discoveries in the Search for Life in the Universe](#)". Recent publications in this area include some by Peter Vickers et al (see below or [here](#)); Vickers [presented](#) in the NASA Goddard series in spring 2023. A related new initiative at Durham, [The Institute for Ascertaining Scientific Consensus](#), launched in 2023, could be of interest in the year/s ahead.

To be continued... for reference...

- Vickers, P et al. 2023. [Confidence of Life Detection: The Problem of Unconceived Alternatives](#). *Astrobiology* July 2023.
- Green, J. et al 2021. [Call for a framework for reporting evidence for life beyond Earth](#). *Nature* 598: 575–579. See also [here](#).
- Meadows, V., H. Graham et al. 2022. [Community Report from the Biosignatures Standards of Evidence Workshop](#).
- [NExSS & NfoLD Standards of Evidence Joint Community Workshop description and talks](#).
- NASA Committee on Astrobiology and Planetary Science review of report, July 2022: [Independent Review of the Community Report from the Biosignature Standards of Evidence Workshop](#) (Jill Tarter participated)

Upcoming (no doubt there are more... please do let me know)

- **IAU/Kavli Symposium, “(Toward) Discovery of Life Beyond Earth and its Impact”**, 15-19 April 2024, Durham UK, [in-person and virtual](#) – involves multiple SETI PC members in the planning and in the [program](#), and will have considerable specific relevance for post-discovery matters
 - NASA Astrobiology Program, virtual workshop, Feb 2024, [“Communicating Discoveries in the Search for Life in the Universe”](#)
 - [AbSciCon 2024](#) Providence, Rhode Island, USA, hybrid in-person/virtual, May 2024
 - [SSOCIA 2024](#) June, Finland
 - Breakthrough Discuss 2024
 - [COSPAR 2024](#)
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3. A few noteworthy publications

(there are more ...)

Chelsea Haramia, 2023. “Understanding the Question of Whether to Message Extraterrestrial Intelligence” in [Routledge Handbook of Social Studies of Outer Space](#), edited by Paola Castaño, Alice Gorman, and Juan Francisco Salazar. Routledge: Oxfordshire, UK.

Michael Bohlander, [Contact with Extraterrestrial Intelligence and Human Law: The Applicability of Rules of War and Human Rights](#). Brill Nijhoff, 2023.

Andreas Anton, M. Schetsche, [Meeting the Alien: An Introduction to Exosociology](#), Springer, 2023. (new English translation)

J. Gertz, Interstellar Diplomacy. Forthcoming 2023 in JBIS. Available: <https://arxiv.org/abs/2308.14917>

Jason Wright, Chelsea Haramia, and Gabriel Swiney. 2022. "[Geopolitical Implications of a Successful SETI Program](#)". *Space Policy*. This paper is a response to Wisian & Traphagan (2020) “The Search for Extraterrestrial Intelligence: A Realpolitik Consideration”, *Space Policy* 52.

Jonathan Jiang et al, 2023. “Message in a Bottle -- An Update to the Golden Record”
<https://arxiv.org/abs/2306.01765>

4. The Protocols: work in progress

In 2022-23, a small task group (Carol Oliver, Kathryn Denning, Michael Garrett, Les Tennen) started work on the *preliminaries* for a revision of the Protocols, building on work done by committee members over the last few years. We welcome your feedback on the IAC paper, which our coauthor Les Tennen just presented in Baku. The abstract is below. [Here is the full manuscript](#). A related 2019 IAC paper, “SETI and Post-Detection: Towards a New Research Roadmap” is [here](#).

Briefly, these are the key points at this stage:

- We have so far only done some very preliminary SETI community consultation, to begin mapping the issues that a revision needs to address. *There will be more opportunities for your input.*
- We are relying on this research community’s collegiality, patience, and expertise as we proceed. Thanks to those who have contributed their perspectives so far.
- The history of the Protocols is underappreciated, and many current members of the Permanent Committee were not present for either the development of the 1989 version or the 2010 version. Anyone who would like to learn more / access resources ... contact me at kdenning@yorku.ca and I’ll share what we have and/or direct you to other experts.
- The initial formulation of the Protocols took years. The 2010 revision also took years. Our situation is more complicated now, due to factors including:
 - the evolution of the science (*e.g.* significant expansion of potential technosignature discovery types, expansion of the technosignatures research community, overlap with biosignatures, increasing potential for serendipitous discovery, etc.)
 - changes in the communications landscape (social media, mis/disinformation, etc) and public <-> science interrelationships
 - the expansion of our Committee (it used to be considerably smaller)
 - accumulated differences of opinion within the community on METI, for example
- We are aiming for:
 - a streamlined formal Declaration of Principles which reflects these evolutions and is more expansive and timeless than the current (2010) Declaration, and which hopefully will not require frequent updating in our era of accelerating change
 - a revised Preamble which makes it clear that our documents do not deal with all possible matters involving potential extraterrestrial life or intelligence
 - auxiliary voluntary Codes of Conduct and Best Practices documents which will be more agile/easily updated, using interdisciplinary expertise (*e.g.* science communication research)
 - groundwork for future discussions for issues which probably can’t be dealt with effectively in a Principles update (*e.g.* METI)
- Stay tuned for more.

The full manuscript (14 pages) for the IAC conference may be found [here](#). (Version: 23 Sept 2023)

Revising the SETI Post-Detection Protocols for the 2020s and Beyond: A Report on Work in Progress.
([IAC-23-A4.2.3 x79618](#), 3 Oct 2023, IAC Baku)

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Abstract: In 1989, the International Academy of Astronautics (IAA) adopted the “Declaration of Principles Concerning Activities Following the Detection of Extraterrestrial Intelligence”, also known as the Search for Extraterrestrial Intelligence (SETI) Post-Detection Protocols. These provided a set of guiding principles to be followed in the event of a putative signal detection, having been developed by the IAA SETI Permanent Committee over a period of several years starting in the late 1980s, and subsequently approved by the International Institute of Space Law (IISL), and endorsed by the Committee on Space Research (COSPAR), the International Astronomical Union (IAU), and the International Astronautical Federation (IAF). A streamlined update to reflect the emergence of the Internet was developed and accepted by the IAA SETI Permanent Committee in 2010 as the “Declaration of Principles Concerning the Conduct of the Search for Extraterrestrial Intelligence”.

Since then, the nature of global communication has been dramatically changed by social media, and multiple SETI ‘false alarms’ have demonstrated the ways in which information moves in our present era. Simultaneously, SETI searches – now also known techno-signature searches – have radically expanded in extent, scope, and power, with substantial search efforts underway at Breakthrough Listen, the SETI Institute, the Chinese Five-hundred-meter Aperture Spherical Telescope (FAST), and many other observatories around the world, between them using many of the world’s most powerful telescopes. Moreover, upcoming next-generation telescopes will allow the detection of molecules indicative of the existence of life in exoplanet atmospheres, conduct high cadence all-sky surveys and be able to detect the equivalent of an airport radar 100 light years away. As a result, the community of scientists engaged in SETI research is expanding significantly. In addition, other methods of detecting exoplanets as potential techno-signature targets are increasing, broadening the community of scientists who may not regard themselves as SETI-related but could potentially uncover data of SETI interest.

In 2022, the IAA SETI Committee decided that a high priority was to update the 2010 IAA Declaration of Principles. A proposal for a small task group with the sole task of updating these Principles was initiated at the end of 2022, and co-chairs put in place to develop that idea. Given rapid changes in SETI/technosignatures research and the dynamic communications landscape, we now envision an eventual combination of timeless Principles which generally preserve the original intentions of the 1989 Protocols and Codes of Conduct and Best Practices documents to provide additional regularly updated guidelines, responsive to changing circumstances. We present a report on the progress on the formation of the task group and its deliberations.